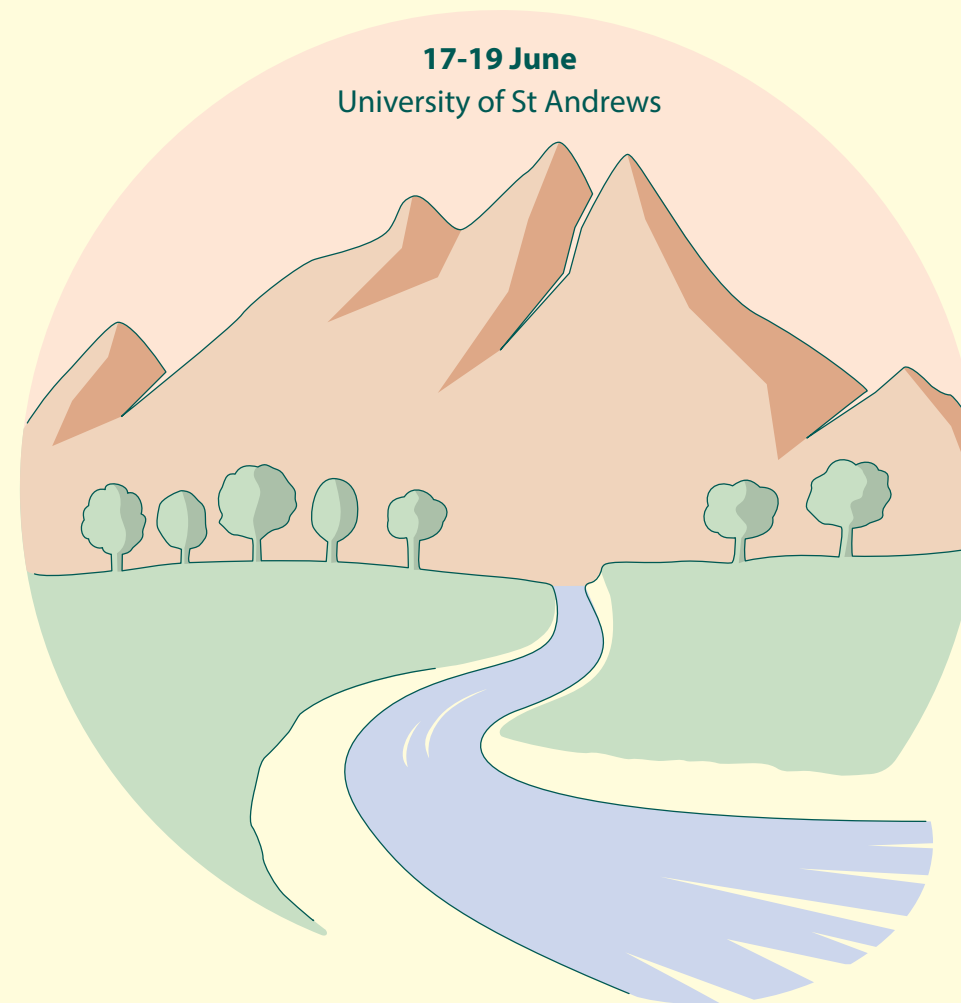




Land-Climate Workshop 2026

17-19 June
University of St Andrews



Financial support from the **Leverhulme Trust** is gratefully acknowledged



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Land-Climate Workshop 2026
17-19 June, University of St Andrews

Organisers:

- Mike Byrne (University of St Andrews)
- Gabi Hegerl (University of Edinburgh)
- Jack Scheff (University of North Carolina at Charlotte)

Accommodation, meals and workshop locations (see map on back page):

- All attendees are staying at Agnes Blackadder Hall (ABH).
- Breakfast will be provided at ABH each day from 7-8.30am. The accommodation is a 20-minute walk from St Mary's Quad, where the workshop will take place.
- All talks and discussion sessions will take place in Divinity Lecture Room 2, St Mary's Quad.
- All lunches and coffee breaks will take place in Parliament Hall, South Street.
- The workshop dinner will take place on Thursday 18th June at 'The Bridge' Restaurant, Rusacks Hotel.

Format of talks: Each speaker is allocated a 30-minute slot. In order to leave plenty of time for questions/discussion, please aim to speak for no more than 20 minutes.

Session 5, part 2 – Fluxes, Hyrology, Drought and Heat

Chair: **David Battisti**

- 11am: **Kirsten Findell** – The importance of daily data in terrestrial hydroclimatic analysis
- 11.30am: **Abigail Swann** – Quantifying the drivers of evapotranspiration over land
- 12pm: **Jack Scheff** – Why is spring longer than fall over midlatitude land in both hemispheres?
- 12.30pm: **Discussion**
- 1pm: **Lunch** – *Parliament Hall*

Session 5, part 3 – Fluxes, Hyrology, Drought and Heat

Chair: **Gabi Hegerl**

- 2pm: **Pierre Gentine** – Roughness or dryness
- 2.30pm: **Hugo Lambert** – Measuring the differences between climate model land surface simulations and FLUXNET observations
- 3pm: **Benjamin Stocker** – How well can we model the response of land-atmosphere CO2 and water fluxes and the response to drought?
- 3.30pm: **Coffee / discussion / wrap-up**

Saturday, 20th June

- By 10am: **Check-out** – *Agnes Blackadder Hall*

Session 4, part 2 – Water Vapour and Land

Chair: Kirsten Findell

- 4pm: **Mike Byrne** – Oceanic-versus-terrestrial influences on land humidity
- 4.30pm: **Marysa Laguë** – Reduced evapotranspiration from plant responses to increased CO2 drives a vertical redistribution of lower tropospheric water vapour
- 5pm: **Yi Zhang** – Power-Law Scaling of Planck Emission with Water Vapor Optical Depth
- 6.30pm: **Workshop dinner** – ‘The Bridge’ restaurant, Rusacks Hotel

Friday, 19th June

- From 7am: **Breakfast** – Agnes Blackadder Hall Dining Room

Session 5, part 1 – Fluxes, Hyrology, Drought and Heat

Chair: David Battisti

- 9am: **Masaki Toda** – Observed deep-tropical land warming since 1980 departs from model and theoretical expectations
- 9.30am: **Hervé Douville** – Land hydrology response to anthropogenic forcing: lessons drawn from the AR6 and new findings
- 10am: **William Boos** – Prediction and processes of monsoon onset for agriculture
- 10.30am: **Coffee** – Parliament Hall

Programme

Tuesday, 16th June

- From 3pm: **Arrival, check-in** – Agnes Blackadder Hall
- 5.30pm: **Informal meetup** – St Andrews Brewing Co

Wednesday, 17th June

- From 7am: **Breakfast** – Agnes Blackadder Hall Dining Room
- 9am: **Registration and welcome remarks from Prof. Tom Brown, Vice-Principal for Research** – St Mary’s Quad, Divinity Lecture 2

Session 1, part 1 – Vegetation, Carbon and Fires

Chair: Jack Scheff

- 9.30am: **David Battisti** – On the non-linear relationship between aridity and (the statistics of) forest fires
- 10am: **Kaighin McColl** – Large overestimation of projected western U.S. wildfire burned forest area with warming
- 10.30am: **Coffee** – Parliament Hall

Session 1, part 2 – Vegetation, Carbon and Fires

Chair: Jack Scheff

- 11am: **René Orth** – Large-scale impacts of water availability and demand on vegetation
- 11.30am: **Ken Takahashi** – AmazClim project: Advancing understanding of coupled forest-atmosphere dynamics in the Amazon through regional Earth system modeling
- 12pm: **Nora Fahrenbach** – Unpacking the potential: How proposed reforestation scenarios shape global and regional climate
- 12.30pm: **Cathy Hohenegger** – Simulating the interactions between the energy, water and carbon cycles at km scale
- 1pm: **Lunch** – Parliament Hall

Session 2, part 1 – Dynamics Over Land

Chair: **Kaighin McColl**

- 2.30pm: **Chris Taylor** – Moist convection and the land surface
3pm: **Steven Sherwood** – The land-ocean contrast in convective intensity – do we know what causes it?

3.30pm: **Coffee** – *Parliament Hall*

Session 2, part #2 – Dynamics Over Land

Chair: **Kaighin McColl**

- 4pm: **Moritz Günther** – Land drives the Walker circulation
4.30pm: **Justin Mankin** – More concentrated precipitation decreases terrestrial water storage

5pm: **Discussion**
5.30-7pm: **Posters, nibbles and drinks** – *Parliament Hall*

Poster presentations:

- **Andrew Chingos** – Amplified Drying of Tropical Land Constrained by Physical Theory
- **Freya Gibbon** – Latitude and strength of westerly winds across a wide range of sea surface temperature gradients
- **Grace Mazaleski** – Comparing Long-term Runoff and Precipitation Trends Worldwide
- **Josh Duffield** – Vertical coupling framework for understanding the spatial variation in the response of temperature extremes to climate change
- **Khushi Dani** – Role of entrainment in shaping the pattern effect
- **Luca Schmidt** – What controls land humidity on hot days in the tropics?
- **Lucie Luecke** – Interpretable machine learning reveals the physical drivers of extreme temperature events in the tropics
- **Lucy Palmer** – Asymmetric Responses of Temperature to Increasing vs Decreasing CO2 Concentrations
- **Roshan Jha** – Understanding Extratropical Moist Heat Extremes: Evidence from Observations, Reanalysis, and Global Storm-Resolving Models
- **Thomas Breitburd** – Learning Compound Climate Extremes: Generative AI for Hot–Dry Event Risk
- **Violet Melcher** – Surface runoff contributions and erosion dynamics in a Scottish agricultural catchment

Thursday, 18th June

From 7am: Breakfast – *Agnes Blackadder Hall Dining Room*

Session 3 – Extremes

Chair: **Cathy Hohenegger**

- 9am: **Erich Fischer** – Record breaking extremes in a warming climate
9.30am: **Quentin Nicolas** – Dry and moist convective upper bounds for near-surface temperatures

10am: **Suqin Duan** – Coherent patterns of land hydroclimate change and some insights on heatwaves

10.30am: **Coffee** – *Parliament Hall*

11am: **Spencer Hill** – ENSO influences on monsoon low-pressure system circulation and extreme rainfall over Indian land
11.30am: **Gabi Hegerl** – High impact events, irreversible change and tipping in land climate: how robust is our understanding?
12pm: **Jacob Zscheischler** – The role of climate model uncertainty and anthropogenic climate change for climate impact assessments

12.30pm: **Discussion**

1pm: **Lunch** – *Parliament Hall*

Session 4, part 1 – Water Vapour and Land

Chair: **Kirsten Findell**

- 2.30pm: **Paul O’Gorman** – Land relative humidity at the Last Glacial Maximum and implications for future climate
3pm: **Lucas Vargas Zeppetello** – What is causing increasing relative humidity over India?

3.30pm: **Coffee** – *Parliament Hall*